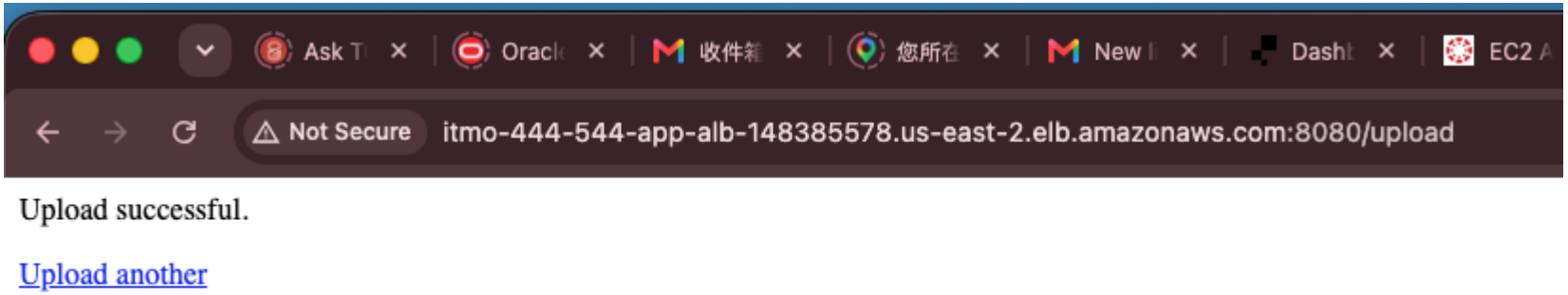
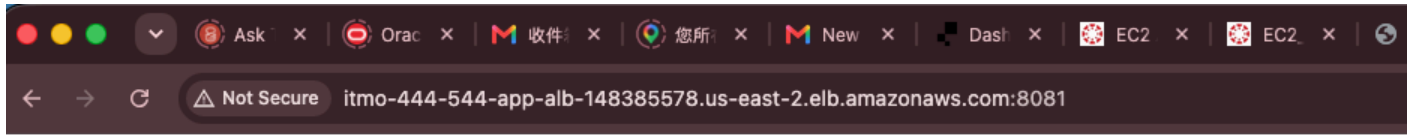
Screenshot of create_app_stack.sh completing, showing both http://<alb-dns>:port/ URLs.

```
root@e4d554fa1453:/ITMD544/project/lab3# ./create_app_stack.sh
== S3 bucket ==
{
    "BucketArn": "arn:aws:s3:::itmo-444-544-jshen25-fileshare",
    "BucketRegion": "us-east-2",
    "AccessPointAlias": false
}
Bucket already exists, skipping.
== Uploader role (PutObject on shared.txt only) ==
Role exists, skipping.
Profile exists, skipping.
== Viewer role (GetObject + ListBucket on shared.txt only) ==
Role exists, skipping.
Profile exists, skipping.
Waiting 15s for IAM propagation...
== Building user-data from local ./ files ==
== Building launch template data ==
== Creating/updating launch templates ==
Template itmo-444-544-uploader-lt exists — publishing a new version
Template itmo-444-544-viewer-lt exists — publishing a new version
== Launching instances from the templates ==
== Looking up the default VPC and subnets in both AZs ==
== ALB security group ==
ALB security group already exists; skipping.
== Tightening the app security group to only accept traffic from the ALB ==
== Target groups ==
Target group itmo-444-544-uploader-tg already exists; skipping.
Target group itmo-444-544-viewer-tg already exists; skipping.
== Application Load Balancer ==
Load balancer itmo-444-544-app-alb already exists; skipping.
== Listeners ==
Listener on port 8080 already exists; skipping.
Listener on port 8081 already exists; skipping.
== Auto Scaling Groups (these launch the instances now, not run-instances) ==
Auto Scaling Group itmo-444-544-uploader-asg already exists; skipping.
Auto Scaling Group itmo-444-544-viewer-asg already exists; skipping.
Done. ALB DNS name: itmo-444-544-app-alb-148385578.us-east-2.elb.amazonaws.com
Uploader: http://itmo-444-544-app-alb-148385578.us-east-2.elb.amazonaws.com:8080/
Viewer:   http://itmo-444-544-app-alb-148385578.us-east-2.elb.amazonaws.com:8081/
Give the target groups 1-2 minutes to report healthy — check with:
  aws elbv2 describe-target-health --target-group-arn arn:aws:elasticloadbalancing:us-east-2:786589010080:targetgroup/itmo-44
4-544-uploader-tg/7c0eac839fed6ec2 --region us-east-2
root@e4d554fa1453:/ITMD544/project/lab3#
```

Screenshots of both apps working through the ALB: uploader upload confirmation and viewer showing content.





Viewer

```
Remote Diagnostic Agent (RDA) 26.2-2026421 for FMW 12c (12.2.1.4)
================================================================================
Patch : 38887908
Date  : 21-Apr-26
Platform Patch for : Generic Platform
Product Patched    : Remote Diagnostic Agent (RDA)
Product Version    : 26.2.26.4.21
================================================================================

This document describes how to install patch # 38887908.

It includes the following sections:

Section 1: FMW Stack Patch Bundle (SPB)
Section 2: Prerequisites
Section 3: Pre-Installation Instructions
Section 4: Installation Instructions
Section 5: Post-Installation Instructions
Section 6: Deinstallation Instructions
Section 7: Additional Notes / Instructions
Section 8: Issues Resolved in Remote Diagnostic Agent (RDA) 26.2.26.4.21
Section 9: Previous Bug Fixes and Enhancements

Section 1: FMW Stack Patch Bundle (SPB)
---------------------------------------
Applying this patch via FMW Stack Patch Bundle (SPB) is recommended.

If SPB can be used, the patch installation instructions in this readme can be
skipped. Instead, follow the install instructions in the Stack Patch Bundle
readme.

Otherwise, this patch can be applied standalone as per the advice given in the
sections below.

Section 2: Prerequisites
------------------------
1. It is recommended to create a backup of ORACLE_HOME and CENTRAL ORACLE
INVENTORY.

With regards to the ORACLE_HOME, this patch will update files in the locations:

- ORACLE_HOME/oracle_common/rda
- ORACLE_HOME/oracle_common/plugins/maven/com/oracle/misc

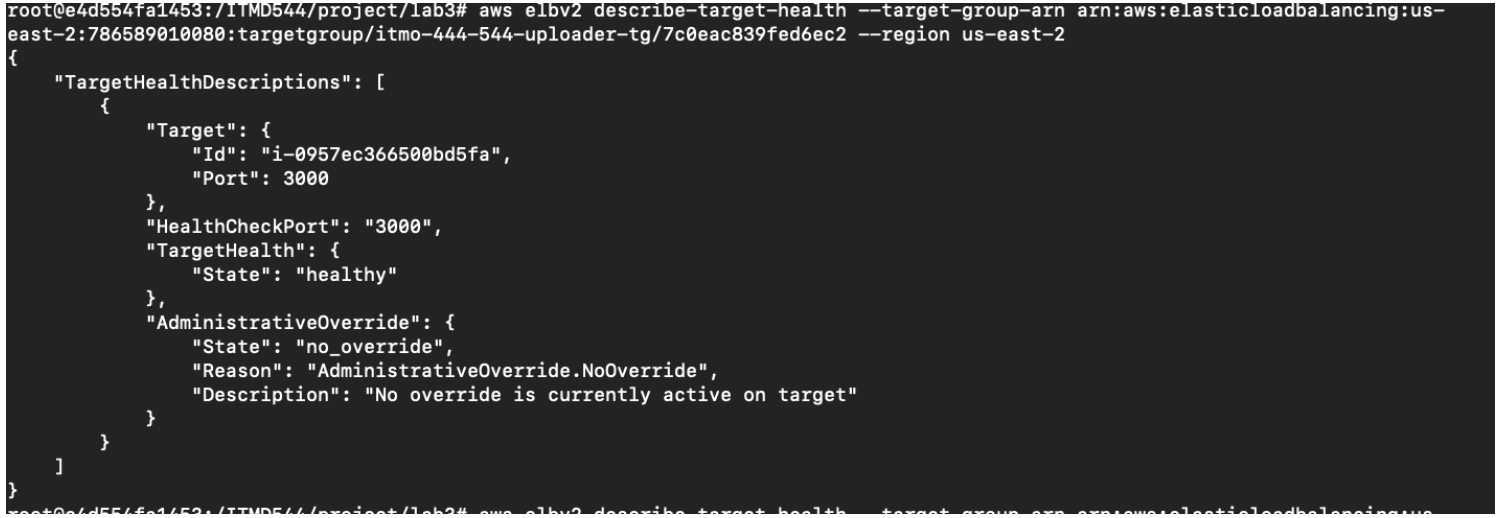
2.  Confirm the OPatch version is 13.9.4.2.13 or higher.

To verify the OPatch version, run the command:

ORACLE_HOME/OPatch/opatch version

To upgrade OPatch to the latest version, download Patch 28186730 and follow the
steps in the Patch 28186730 README.
```

Screenshot of aws elbv2 describe-target-health showing both target groups healthy.



The CSV poll log and console output from a chaos_test.py run that passed its SLO, for at least one tier.

```
root@e4d554fa1453:/ITMD544/project/lab3#
root@e4d554fa1453:/ITMD544/project/lab3# python3 testing/chaos_test.py --tier uploader
Target: uploader tier at http://itmo-444-544-app-alb-148385578.us-east-2.elb.amazonaws.com:8080/
Confirming the endpoint is healthy before we start...
Terminating i-0f60844e88aa31d7a via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_uploader_36057.csv
Recovery time: 87.1s (SLO: 180s)
PASS: recovery time met the SLO.
root@e4d554fa1453:/ITMD544/project/lab3# python3 testing/chaos_test.py  --tier  viewer  --slo-seconds   120
Target: viewer tier at http://itmo-444-544-app-alb-148385578.us-east-2.elb.amazonaws.com:8081/
Confirming the endpoint is healthy before we start...
Terminating i-0e0092581be1b6927 via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_viewer_36682.csv
Recovery time: 84.0s (SLO: 120s)
PASS: recovery time met the SLO.
root@e4d554fa1453:/ITMD544/project/lab3#
```

```
root@e4d554fa1453:/ITMD544/project/lab3# more chaos_test_uploader_35206.csv
elapsed_seconds,is_up
0.0,True
2.11,False
0.0,False
2.11,False
4.22,False
6.34,False
8.45,False
10.56,False
12.67,False
14.79,False
16.91,False
19.02,False
21.13,False
23.24,False
25.35,False
27.47,False
29.58,False
31.69,False
33.8,False
35.91,False
38.04,False
40.16,False
42.27,False
44.39,False
46.5,False
48.61,False
50.72,False
52.84,False
54.95,False
```

```
511.68,False
513.79,False
515.9,False
518.02,False
520.13,False
522.24,False
524.35,False
526.46,False
528.58,False
530.68,False
532.79,False
534.91,False
537.03,False
539.14,False
541.26,False
543.38,False
545.5,False
547.62,False
549.73,False
551.84,False
553.96,False
556.07,False
558.18,False
560.31,False
562.41,False
564.52,False
566.64,False
568.75,False
570.87,False
572.99,False
575.09,False
577.21,False
579.33,False
581.45,False
583.56,False
585.67,False
```

The CSV poll log and console output from a run that failed a tightened SLO.

```
root@e4d554fa1453:/ITMD544/project/lab3# python3 testing/chaos_test.py --tier uploader
Target: uploader tier at http://itmo-444-544-app-alb-148385578.us-east-2.elb.amazonaws.com:8080/
Confirming the endpoint is healthy before we start...
Terminating i-0957ec366500bd5fa via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_uploader_35206.csv
FAIL: endpoint did not recover within max-wait (600s).
root@e4d554fa1453:/ITMD544/project/lab3# aws autoscaling describe-scaling-activities \
  --auto-scaling-group-name itmo-444-544-uploader-asg \
  --max-items 5 \
  --query "Activities[].{Time:StartTime,Status:StatusCode,Description:Description,Message:StatusMessage}" \
  --output json \
  --region us-east-2
[
    {
        "Time": "2026-10-18T03:04:54.235000+00:00",
        "Status": "Cancelled",
        "Description": "Launching a new EC2 instance.  Status Reason: Invalid launch template: When a network interface is provided, the security groups must be a part of it.",
        "Message": "Invalid launch template: When a network interface is provided, the security groups must be a part of it."
    },
    {
        "Time": "2026-10-18T02:59:31.632000+00:00",
        "Status": "Cancelled",
        "Description": "Launching a new EC2 instance.  Status Reason: Invalid launch template: When a network interface is provided, the security groups must be a part of it.",
        "Message": "Invalid launch template: When a network interface is provided, the security groups must be a part of it."
    },
    {
        "Time": "2026-10-18T02:59:24.031000+00:00",
        "Status": "Successful",
        "Description": "Terminating EC2 instance: i-0957ec366500bd5fa",
        "Message": null
    },
    {
```

```
-rw-r--r-- 1 root root      515 Oct 18 03:25 chaos_test_viewer_36682.csv
root@e4d554fa1453:/ITMD544/project/lab3# cat chaos_test_uploader_36057.csv
elapsed_seconds,is_up
0.0,True
2.11,False
0.0,False
2.11,False
4.21,False
6.32,False
8.43,False
10.54,False
12.65,False
14.76,False
19.83,False
24.9,False
29.97,False
32.08,False
34.2,False
36.32,False
38.43,False
40.55,False
42.66,False
44.78,False
46.9,False
49.02,False
51.13,False
53.23,False
55.35,False
57.47,False
59.59,False
61.7,False
63.82,False
65.93,False
68.04,False
```

One paragraph explaining, using your own recovery-time numbers, what dominates the recovery window: health-check timing, boot/npm install time, or something else.

In my uploader test, the measured recovery time was 87.1 seconds, which passed the 180-second SLO. Starting the replacement EC2 instance and installing Node.js and npm packages likely took most of this time. The health checks also added delay because they ran every 10 seconds and required two successful checks. However, I did not measure each stage separately, so I cannot confirm which stage took the longest. My earlier test failed after 600 seconds because an invalid launch template prevented a replacement instance from starting. This shows that configuration errors can prevent recovery completely, while startup time and health checks affect recovery when the configuration is correct. The 87.1 seconds was measured from the start of waiting for recovery, not from the termination request.

One paragraph explaining what changed in create_app_stack.sh and delete_app_stack.sh between the previous lab and this one—specifically, what replaced the old run-instances call and why.

In the previous lab, create_app_stack.sh used aws ec2 run-instances to launch the uploader and viewer directly. In this lab, Auto Scaling Groups replaced those calls and used launch templates to create instances. Each group maintains one instance and replaces it if it is terminated or becomes unhealthy. The script also creates an ALB, target groups, and listeners so users can access the apps through stable URLs when instances change. In delete_app_stack.sh, cleanup now deletes the Auto Scaling Groups and their instances first, then removes the ALB, target groups, and ALB security group before cleaning up IAM resources and the S3 bucket.

Screenshot of delete_app_stack.sh completing with no errors.

```
root@e4d554fa1453:/ITMD544/project/lab3# ./delete_app_stack.sh
== Deleting Auto Scaling Groups (this also terminates their instances) ==
Waiting for both Auto Scaling Groups to be gone...
itmo-444-544-viewer-asg deleted.
== Deleting the load balancer ==
== Deleting target groups ==
== Restoring the app security group to its previous-lab state ==
== Deleting the ALB security group ==
== Removing IAM instance profiles and roles ==
== Emptying and deleting S3 bucket ==
delete: s3://itmo-444-544-jshen25-fileshare/shared.txt
Cleanup complete.
```